Early Dynamics Predict Model Performance

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1. Introduction

As large language models (LMs) continue to improve with scale, it is increasingly and prohibitively expensive to explore modeling decisions at full scale. Simultaneously, modeling choices including pretraining data recipes **cite**, learning rate schedules **cite**, data delivery schedule **cite**, ... (**fix this list**) have been shown to significantly impact final performance.

Then, it has become best practice to combine small scale experiments with scaling laws to predict the best design for full scale performance (Li et al., 2024; Kaplan et al., 2020; Hoffmann et al., 2022; Choshen et al., 2024), and even for downstream performance (Gadre et al., 2024; Bhagia et al., 2024), at a fraction of the cost.

However, recent work has suggested that these approaches based on scaling laws are not always accurate predictors (Li et al., 2025; Lourie et al., 2025).

We propose an alternative method of reducing the cost of selecting the best model recipe for large scale training by using metrics from the first period of training to predict later performance. We obtain a XXX% accuracy of predicting the performance ranking of models trained across DataDecide (Magnusson et al., 2025) recipes on IID and OOD datasets and the OLMES (Gu et al., 2024) set of downstream tasks at the final epoch of training, using metrics from just the first 10% of training.

2. Related Work

- Loss curve prediction (Brandfonbrener et al., 2024)
- Predicting performance from early metrics like training speed for NAS (Ru et al., 2021)
- Loss landscape metrics

3. Methods

3.1. Models and Datasets

We consider the data recipes introduced in the DataDecide benchmark (Magnusson et al., 2025), described in Table 1. Across all data recipes, for each of 14 model sizes, all checkpoints throughout training have been evaluated on the following validation splits to track the perplexity evolution:

- wikitext_103-validation
- pile-validation
- m2d2_s2orc-validation
- ice-validation
- dolma_wiki-validation
- dolma_stack-validation
- dolma_reddit-validation
- dolma_pes2o-validation
- dolma_common-crawl-validation
- dolma_books-validation
- c4_en-validation

They've additionally evaluated on the downstream tasks included in the OLMES (Gu et al., 2024) benchmark: MMLU (Hendrycks et al., 2021), HellaSwag (Zellers et al., 2019), ARC Challenge (Clark et al., 2018), ARC Easy (Clark et al., 2018), PIQA (Bisk et al., 2020), CommonsenseQA (Talmor et al., 2019),SocialIQA (Sap et al., 2019), OpenBookQA (Mihaylov et al., 2018), BoolQ (Clark et al., 2019), and WinoGrande (Sakaguchi et al., 2020).

These evaluation curves provide the metrics used to predict final model performance.

3.2. Problem Setup

Given M model training setups, perform training for the first S_0 steps. Use the metrics from these initial steps to predict the performance at step S.

We predict downstream performance from a model at step S via the CORRECT PROB continuous proxy metric, the average probabilities of the correct continuations, which was shown by Magnusson et al. (2025) to provides a larger spread of performances across models per recipe and reduced noise in evaluation for many tasks at small scales, in contrast to discrete metrics like accuracy which Schaeffer

Source / Recipe	Description
Dolma1.7 *Original, No code, No math/code, No Reddit, No Flan*	A 2.3T-token corpus (Dolma 1.7 Soldaini et al., 2024) sampling common LM sources for open research. We ablate code, math/code, Reddit, or Flan subsets.
Dolma1.6++ *Original*	Dolma 1.6 plus additional sources from Dolma 1.7: RedPajama's arxiv subset, openwebmath, algebraic stack, flan, starcoder, falcon.
C4 *Original*	The C4 dataset (Raffel et al., 2019) as prepared in Dolma 1.7, heuristically filtered from the April 2019 Common Crawl.
FineWeb-Pro *Original*	The FineWeb Pro corpus (**?**), featuring model-driven data cleaning on FineWeb.
FineWeb-Edu *Original*	The deduplicated FineWeb-Edu subset of SmolLM-Corpus (Ben Allal et al., 2024), focused on educational web pages.
Falcon *Original*	The Falcon RefinedWeb corpus (Penedo et al., 2023) in Dolma 1.7, derived from Common Crawl through June 2023 and more aggressively filtered/deduplicated than C4.
Falcon+CC *Original, QC 10%, QC 20%, QC Orig 10%, QC Tulu 10%*	Falcon and Dolma 1.7's Common Crawl. We quality filter to top 10% or 20% documents with reproduced or original (Li et al., 2024) filter or retrain filter on pre-release version of Tulu-v3 (Lambert et al., 2024).
DCLM-Baseline *Original, QC 7% FW2, QC 7% FW3, QC FW 3%, QC FW 10%, QC 10%, QC 20%*	A SOTA Common Crawl corpus using best ablated deduplication, cleaning heuristics, and quality filter. We quality filter to top 7% of DCLM classified documents and further take 2+ or 3+ scores with FineWeb-edu classifier; or filter to top 3% or 10% with FineWeb-edu classifier; or take top 10% or 20% with reproduced DCLM classifier.
$\lambda\%$ **DCLM-Baseline** + $1-\lambda\%$ **Dolma1.7**	Fractional combinations of Dolma1.7 and DCLM-Baseline mixing different proportions of the two datasets for $\lambda \in \{25\%, 50\%, 75\%\}$.

Table 1. The DataDecide recipes, each released with all checkpoints for models of 14 sizes (4M, 6M, 8M, 10M, 14M, 16M, 20M, 60M, 90M, 150M, 300M, 530M, 750M, 1B) for 3 seeds each. *Reproduced directly from DataDecide paper (Magnusson et al., 2025).*

et al. (2023) find to result in jumps in performance across scales.

Additionally we predict the ranking of the models at step S.

3.3. Evaluation

We follow Magnusson et al. (2025) in evaluating absolute and relative **prediction error** on the CORRECT PROB proxy metric as $(\frac{|\text{predicted}-\text{actual}|}{\text{actual}} \times 100\%)$ and $(|\text{predicted} - \text{actual}| \times 100\%)$ respectively.

Additionally we report **decision accuracy** to capture the quality of the ranking as:

$$\frac{1}{|\mathcal{P}|} \sum_{(A,B)\in\mathcal{P}} \mathbb{I}\big(\text{sign}(\hat{y}_A - \hat{y}_B) = \text{sign}(y_A - y_B)\big) \quad (1)$$

TODO Add information about rescaling to [0,1].

3.4. Prediction Methods

- Start simple: Linear regression
- Possibly more complex: Gradient Boosted Decision Trees
- Interpretable: Gaussian Processes

4. Experiments

For each experiment we explore generalization across data recipes and across model scales.

For generalization across data recipe we train one predictor across all model sizes and validate our method using leave-one-out cross validation on both the single recipe and recipe-family level, meaning we train on the evaluation results from all but one recipe (or family) and evaluate our predictive performance on the remaining recipe (or family). Our final metric is the average across all held-out sets.

For generalization across model scales we train one predictor across all avilable recipes and validate with expanding window validation, starting by training on models of size 4M - 20M and validating on all larger models and then repeating the process as we expand the training set.

4.1. In Domain Prediction

The first experiment considers whether the performance on any task at timestep S can be predicted using the performance on the same task from the first S_0 steps.

Correlation Results. We first explore correlation.

Prediction Results. For the In Domain setting we validate our methods using leave-one-out cross validation on both the single recipe and recipe-family level. Then, we train on the evaluations from all but one recipe (or family) and evaluate our predictive performance on the remaining. Our final metric is the average across all held-out sets.

4.2. All to All Prediction

The next experiment considers whether the performance of a single representative downstream task, MMLUcan be predicted using all available evaluations for the first S_0 steps.

4.3. Prediction Results

Then we want to get prediction results, so we need to decide how to determine the held-out sets (larger scales, random datasets?).

Then we need to decide what we're trying to predict from what:

- early - late
- ppl - downstream
- downstream - ppl
- early ppl + downstream - OOD ppl+downstream late
- all early - all late simultaneously

5. Future Work

5.1. Finetune Performance

Predict finetune performance from metrics across pretraining and the early finetune metrics. Consider SFT and DPO datasets and standard evaluation sets (eg. for math, coding, instruction following settings).

5.2. Extended Early Training Metrics

Include metrics that capture more precise aspects of the loss landscape and training dynamics potentially including:

- **Curvature:** via the max eigenvalues of the Hessian and an approximation of the Hessian trace. Lower curvature is potentially related to better optimization dynamics and finetuning performance.
- **Feature Stability:** less forgetting but potentially also less plasticity.
- **Gradient and Activation Variance:** Potentially less interference between knowledge during acquisition.

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Model name	Batch size	Hidden dim.	LR	Model size	Heads	Layers	Training steps	Tokens trained
4M	32	64	1.4e-02	3.7M	8	8	5,725	0.4B
6M	32	96	1.2e-02	6.0M	8	8	9,182	0.6B
8M	32	128	1.1e-02	8.5M	8	8	13,039	0.9B
10M	32	144	1.0e-02	9.9M	8	8	15,117	1.0B
14M	32	192	9.2e-03	14.4M	8	8	21,953	1.4B
16M	32	208	8.9e-03	16.0M	8	8	24,432	1.6B
20M	64	192	8.4e-03	19.1M	8	16	14,584	1.9B
60M	96	384	5.8e-03	57.1M	12	16	29,042	5.7B
90M	160	528	4.9e-03	97.9M	12	16	29,901	9.8B
150M	192	768	4.2e-03	151.9M	12	12	38,157	15.0B
300M	320	1,024	3.3e-03	320.0M	16	16	45,787	30.0B
530M	448	1,344	2.8e-03	530.1M	16	16	57,786	53.0B
750M	576	1,536	2.5e-03	681.3M	16	16	63,589	75.0B
1B	704	2,048	2.1e-03	1176.8M	16	16	69,369	100.0B

Table 2. DATADECIDE uses OLMo's *model ladder* (Groeneveld et al., 2024; OLMo et al., 2025; Bhagia et al., 2024) to programmatically create configurations for 14 model sizes with hyperparameters determined by heuristics in Porian et al. (2024). All models have sequence length of 2024 and MLP ratio of 8. Each configuration is pretrained over 25 data recipes (Table 1). Each recipe and configuration is also trained for 3 random seeds where model sizes $<$ 1B are stopped early at 25% of the compute used to train the 1B model for all but the default seed. Model size is number of non-embedding parameters. Batch size is the number of sequences per batch. *Reproduced directly from DataDecide paper (Magnusson et al., 2025)*

A. DataDecide Model Training Details